

# Tutorial: Sketching a Jack-o'-Lantern with KodaCAD's 2D Tools

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This one is primarily about the 2D sketch toolbar -- construction lines, construction circles, and the geometry tools that turn a layout into a real profile. We'll draw a jack-o'-lantern face on a workplane, one feature at a time, then -- as a bonus, once the sketching itself is solid -- use that same face to carve an actual pumpkin.

## What you'll need

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A fresh (empty) KodaCAD session. No file import required -- everything here is built entirely from scratch.

## What this exercises

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Nearly the full 2D sketch toolbar in one continuous drawing: Parallel Construction Line, Angled Construction Line, Construction Line by 2 Points, Construction Circle, Constr Circle by 3 Points, Slot, Polyline, Arc by 3 Points, Arc: Center + 2 Points, and both Delete Construction Element and Delete Geometry Element -- deliberately using more than one tool to build equivalent geometry (the two eyes), so the same result is reachable more than one way. The second half builds on the Bottle tutorial's pull-then-heavily-fillet technique for a rounded, non-revolved form, then shells the form and uses section view to verify the shell operation, then moves the workplane with the face features into position on the front of the pumpkin to demonstrate Pull's multi-profile capability to remove material from the pumkin.

We'll be drawing with two kinds of lines throughout:

- **Construction lines** -- the dotted magenta lines used to lay out the drawing accurately.
- **Geometry lines** -- the heavy, solid lines that make up the actual **profiles** used to create or modify 3D shapes.

The goal of every profile below is a simple, closed loop of geometry lines -- nothing extraneous, no open gaps. Get that right and the CAD kernel turns it into a **wire**, ready to be pulled. Get it wrong and the kernel won't accept it.

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## Step 1 -- Make the nose

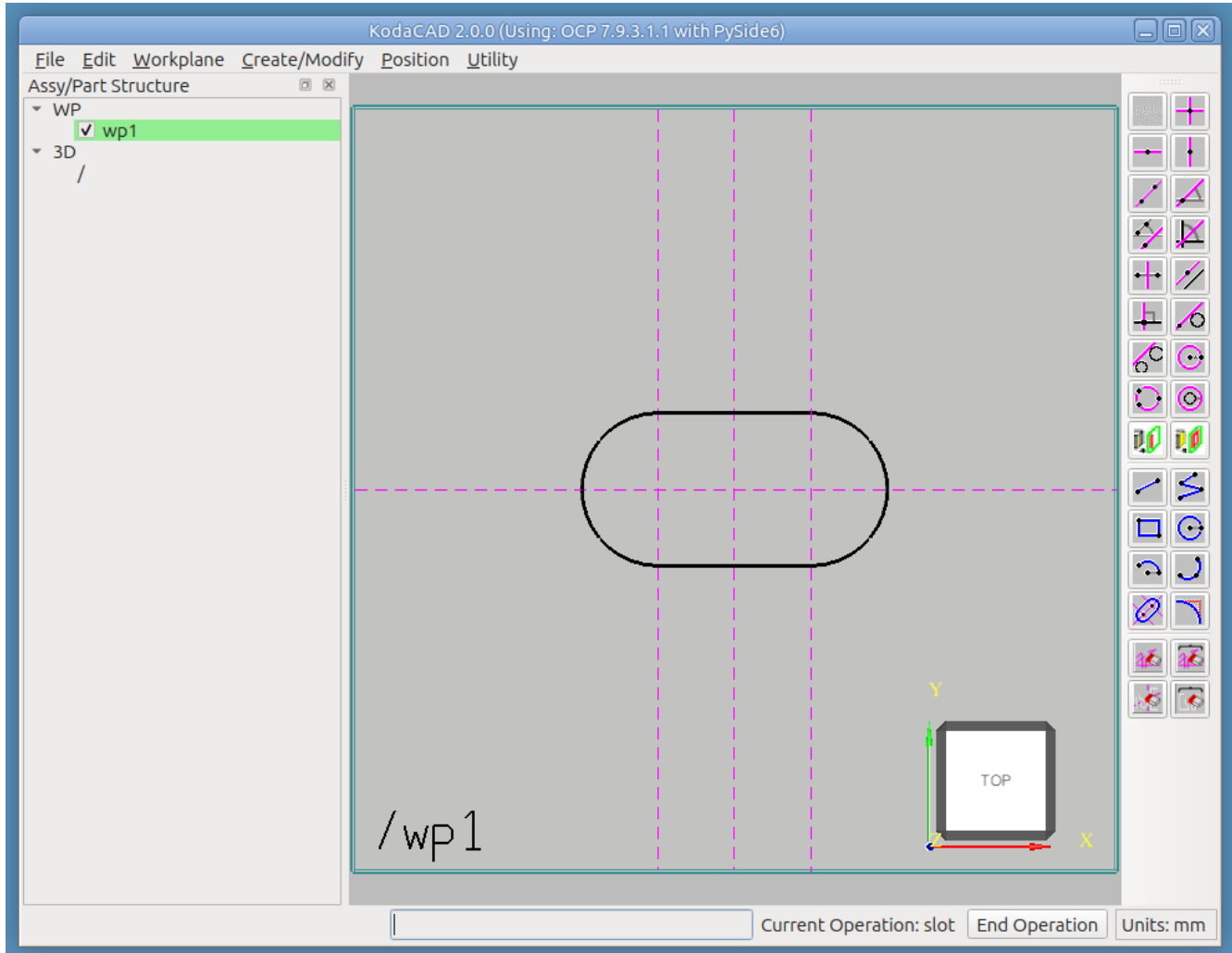
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This first workplane is just for practice -- feel free to make mistakes and try out both the construction and geometry tools freely.

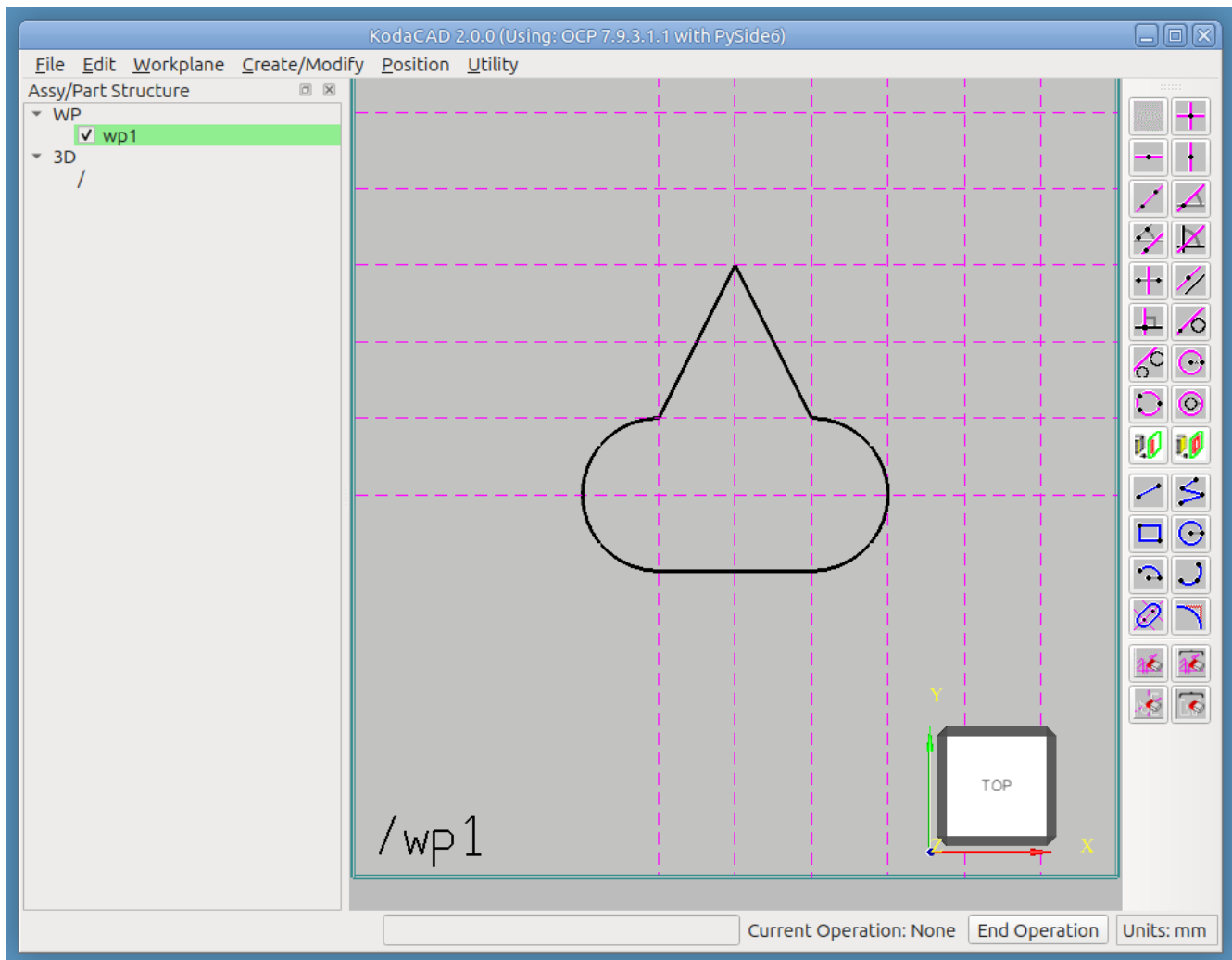
- Start a new KodaCAD session and create a workplane at the origin on the XY plane. We'll use a 20 mm grid of construction lines throughout, to keep the layout accurate without cluttering it with dimensions.
- Click the **Parallel Construction Line** tool, enter 20 , then click the vertical construction line, move the cursor to one side, and click to place a second vertical line 20 mm away. Repeat on the other side of

the original line.

- Use the **Slot** tool to click the two intersection points to the left and right of the origin, then enter a slot width of 40 .



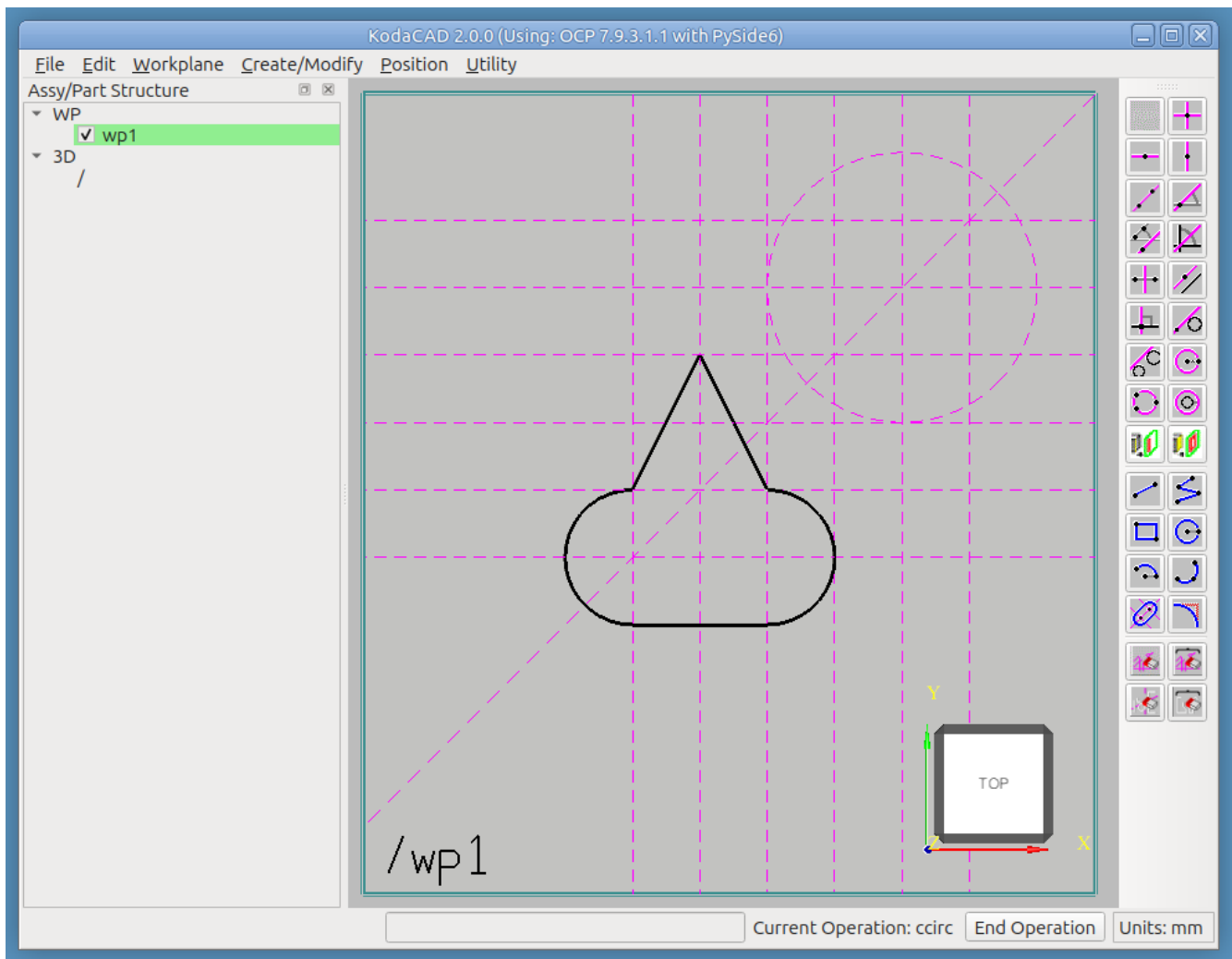
- Add some more grid lines above and to the right of the slot.
- Finish the nose with the **Polyline** tool: two diagonal lines meeting at a point above the slot, as shown below. Then use **Delete Geometry Element** to remove the slot's own upper straight edge, since the two new diagonals replace it.



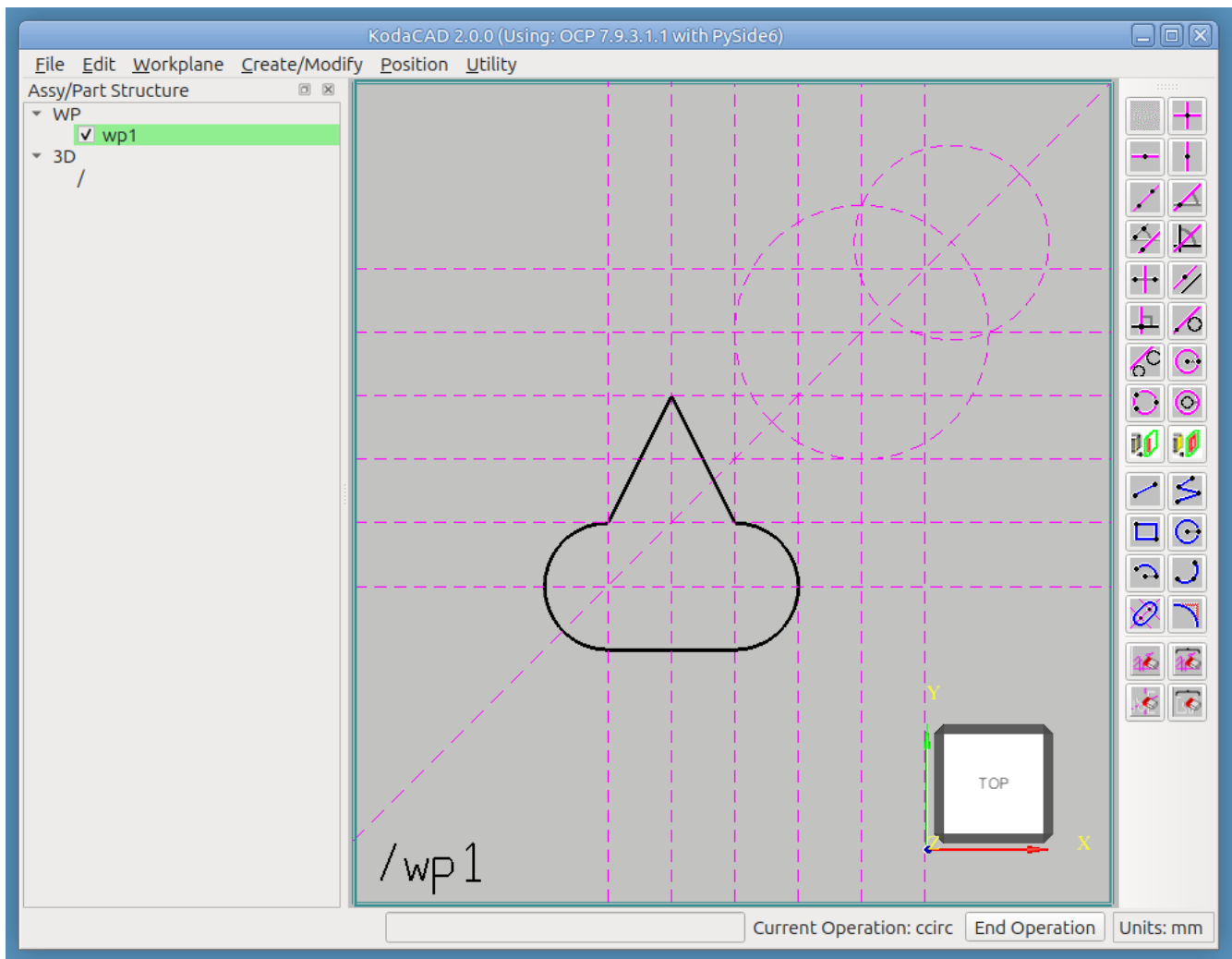
*Exercises: Parallel Construction Line for a symmetric grid; Slot; Polyline; Delete Geometry Element – and, implicitly, that a profile can be assembled from more than one tool's own geometry (the slot's two remaining sides plus the polyline's two new ones) as long as the final loop closes cleanly.*

## Step 2 -- Make the eyes

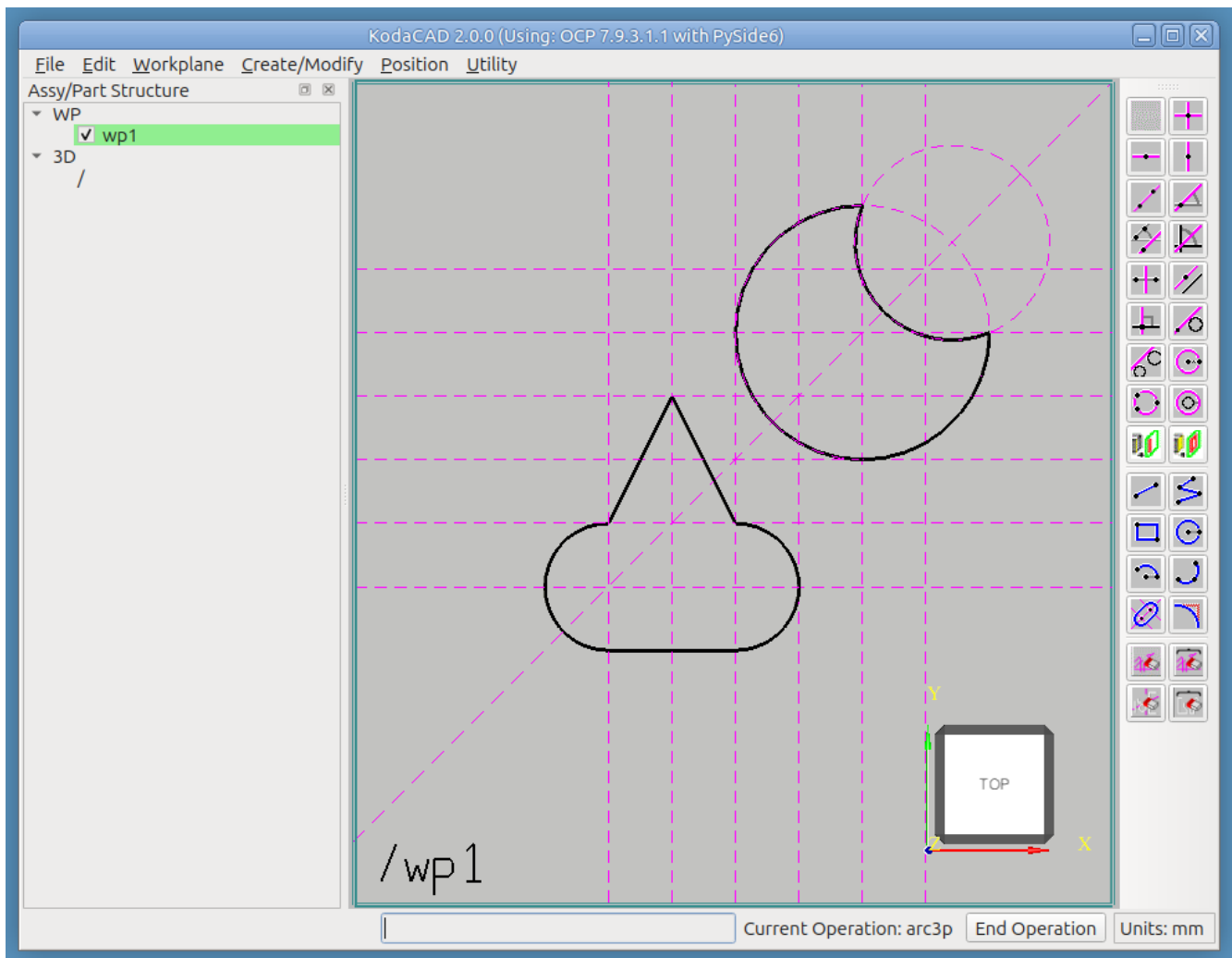
- Use **Construction Line by 2 Points** to add a 45-degree angled construction line, and **Construction Circle** to add a circle above and to the right of the nose -- this is the left eye's location.



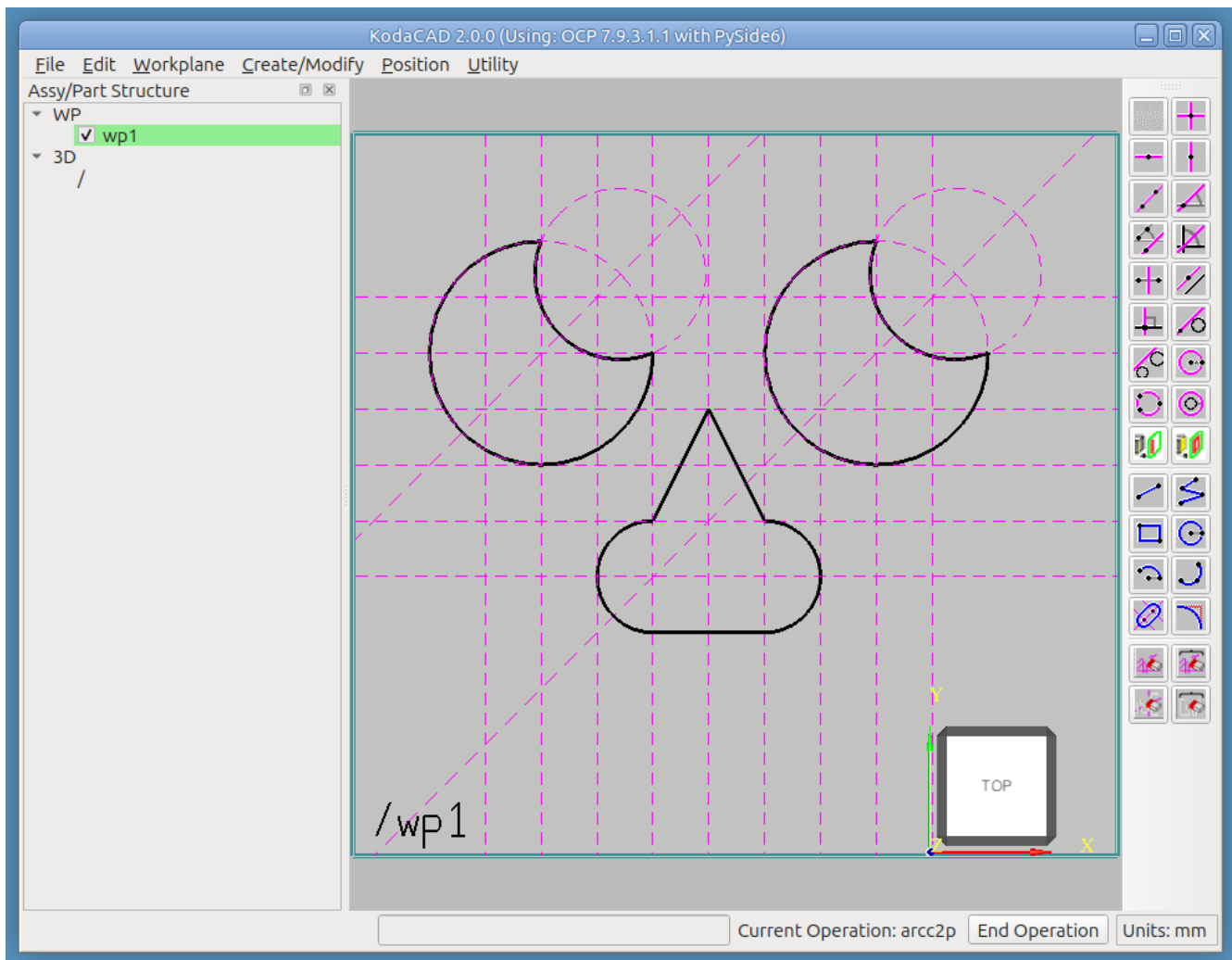
- Using the same Construction Circle tool, add a second circle. Click first at the intersection of the first circle and the 45-degree line to set its center, then click again at the intersection of the first circle and either the horizontal or vertical construction line through its own center.



- Use **Arc by 3 Points** to place the two arcs shown below. For each arc, click the two end points first, then a point where the arc crosses the 45-degree construction line.



- Repeat all of this for the right eye, but try different tools this time, to see that the same shape is reachable more than one way:
  - **Angled Construction Line** instead of Construction Line by 2 Points, for the 45-degree line.
  - **Constr Circle by 3 Points** instead of Construction Circle, for the first circle.
  - **Arc: Center + 2 Points** instead of Arc by 3 Points, for the two arcs. This one takes a little practice – after clicking the center, the two end points have to be clicked in CCW order (the standard positive direction for measuring arcs). If a click goes wrong, **Delete Geometry Element** removes just the mistake, without disturbing anything else.

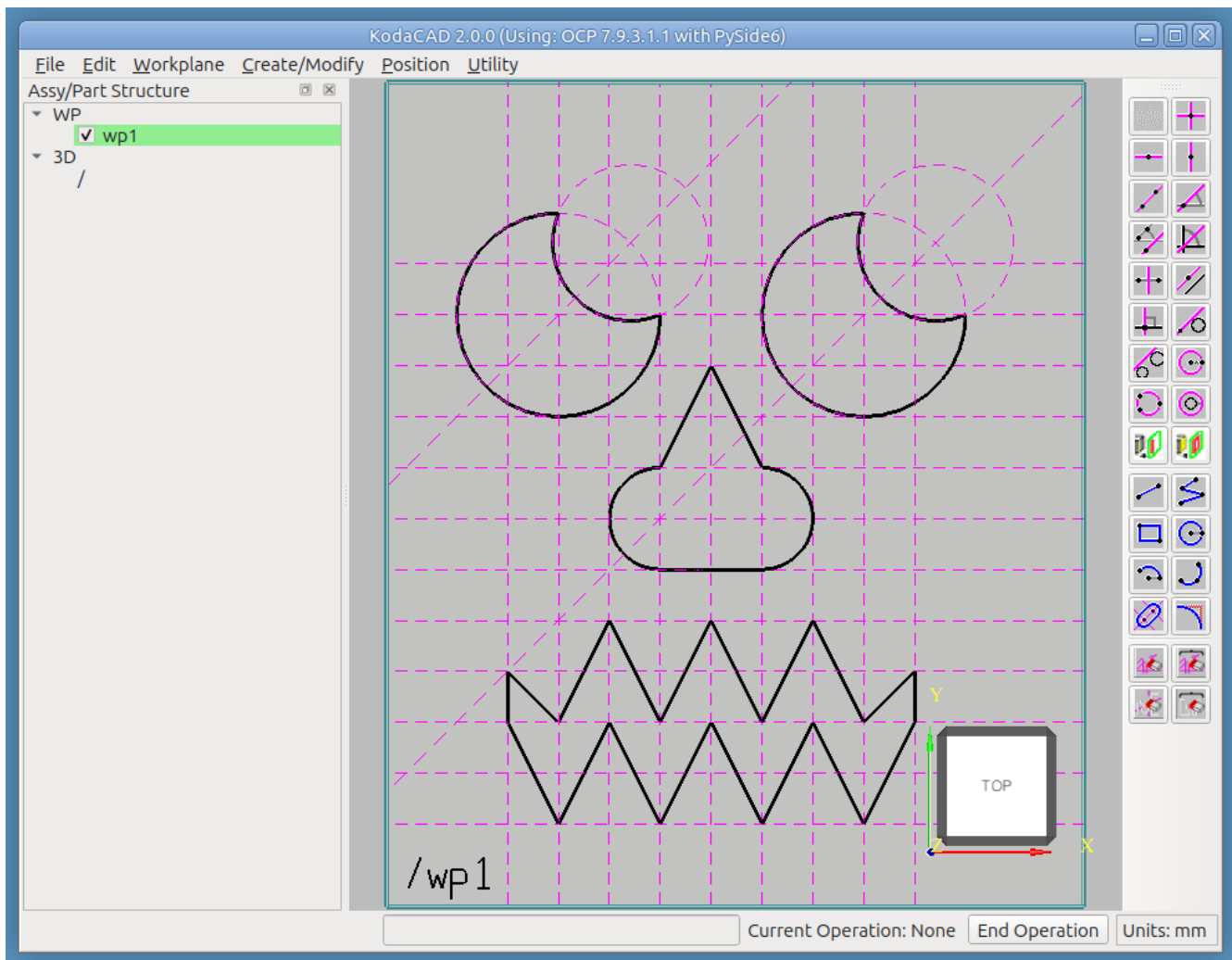


Mistakes on the workplane itself are best undone with **Delete Construction Element** or **Delete Geometry Element**, picking off just the one line that's wrong. The Edit menu's Undo/Redo are for undoing changes to the 3D model, not individual sketch lines.

*Exercises: two complete, independent tool paths for equivalent geometry -- Construction Line by 2 Points + Construction Circle + Arc by 3 Points for the left eye, versus Angled Construction Line + Constr Circle by 3 Points + Arc: Center + 2 Points for the right -- plus Delete Construction Element and Delete Geometry Element for correcting mistakes along the way.*

## Step 3 -- Make the mouth

- Expand the grid below the nose in preparation for the mouth.
- Use **Polyline** to click through every point of the mouth in sequence.



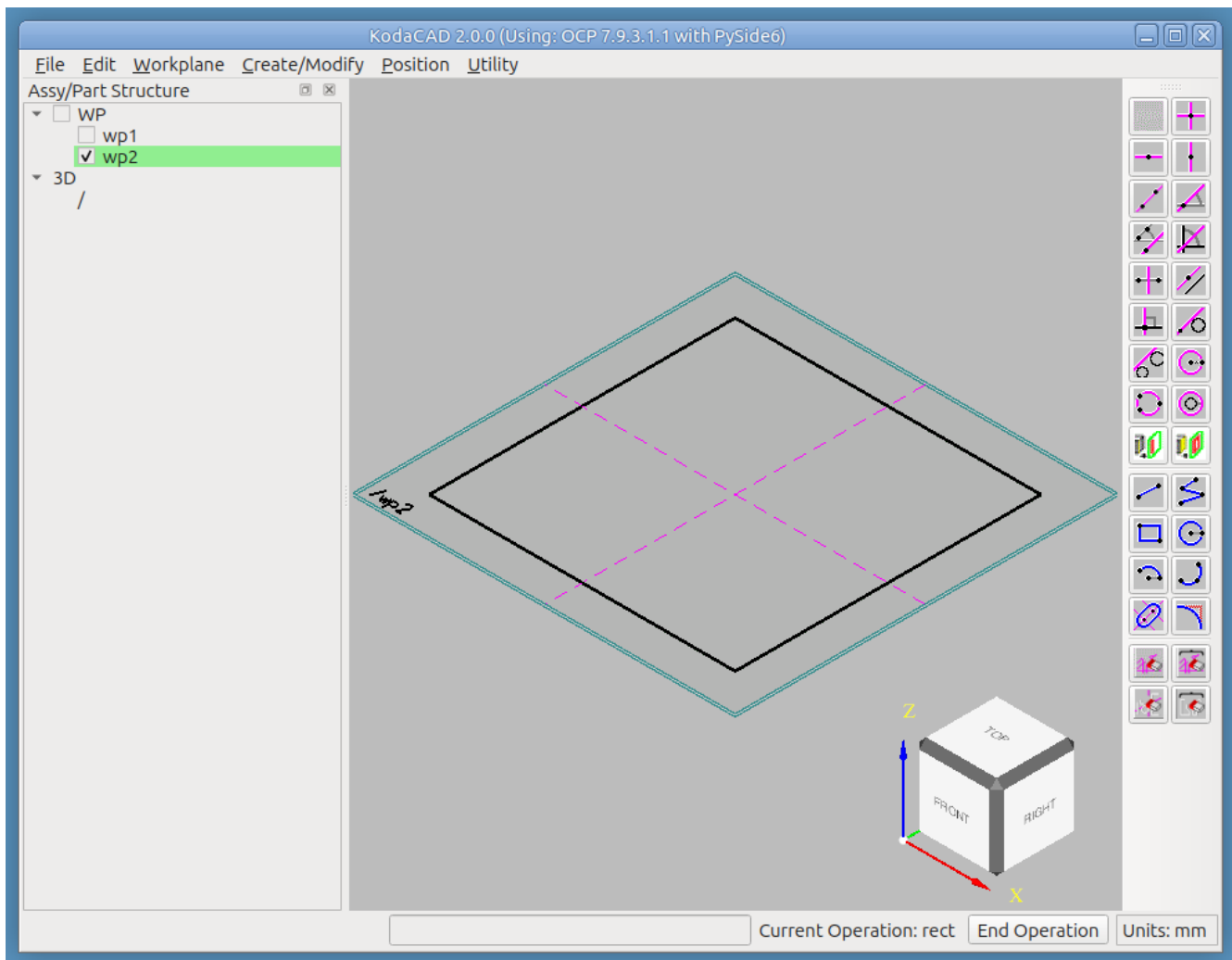
*Exercises: Polyline for a longer, more irregular closed profile than the nose's simpler one.*

## Building the pumpkin

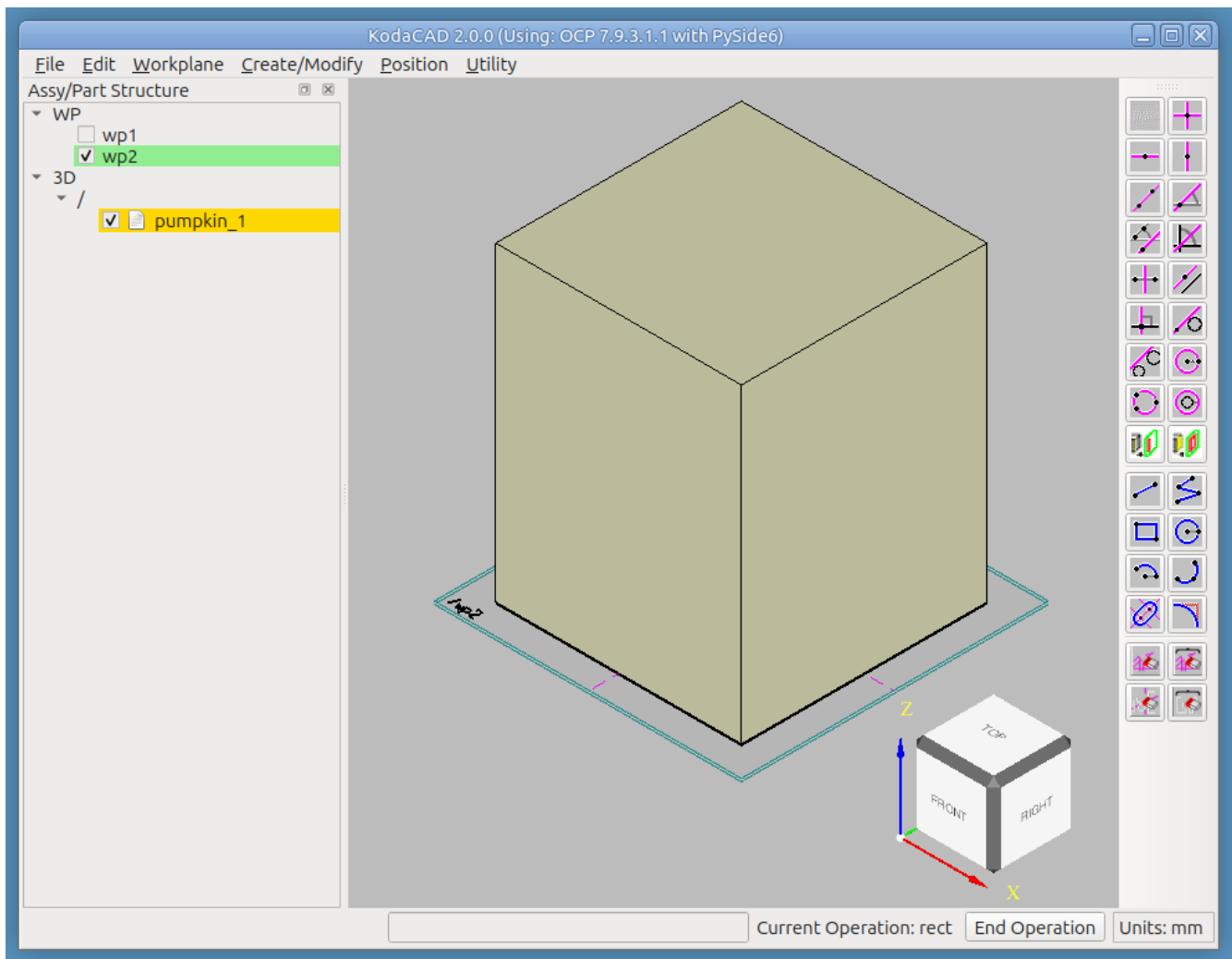
The face is done. From here, we build an actual pumpkin and carve the face into it.

### Step 4 -- Pull a square profile

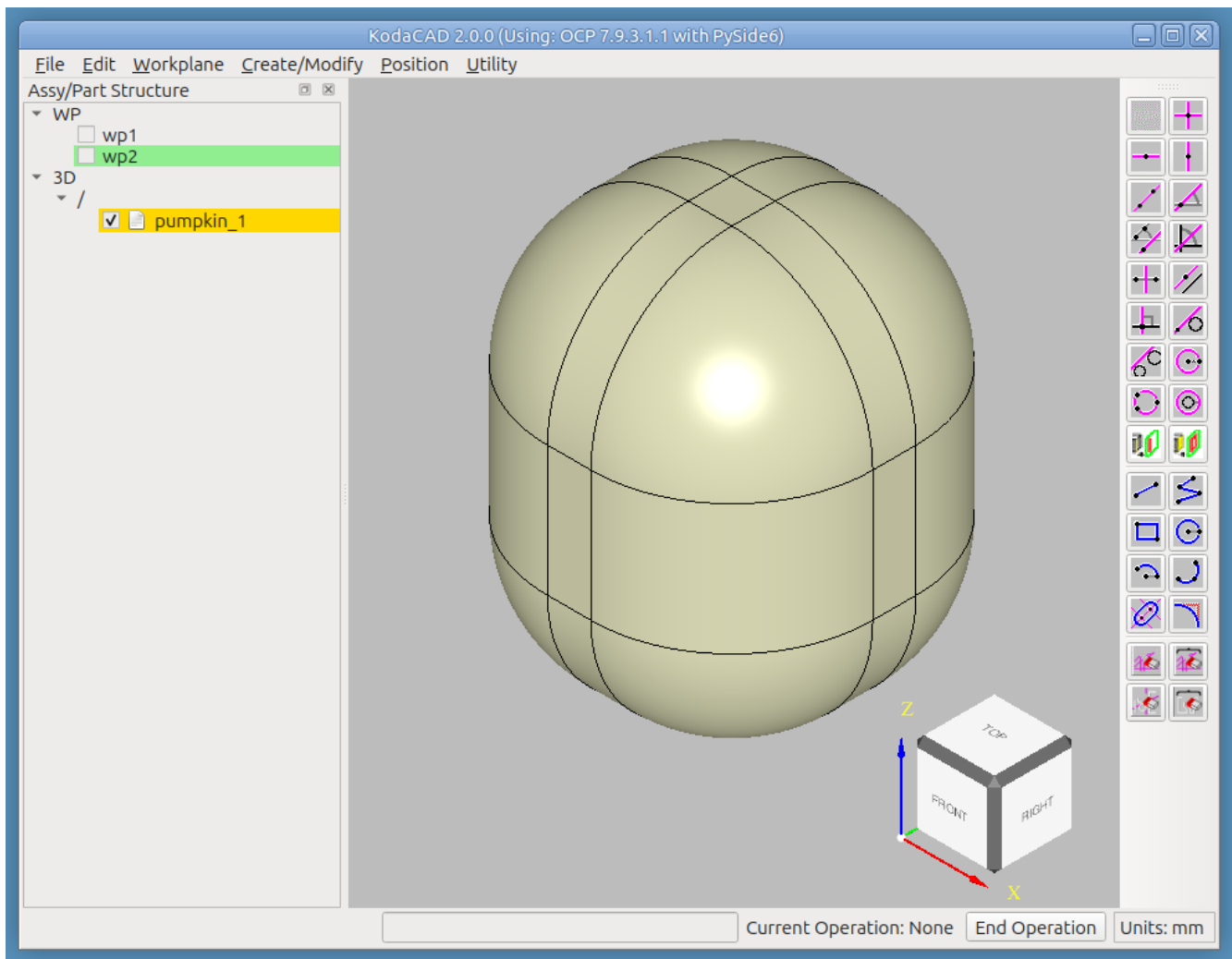
- Hide wp1. We'll use it later.
- Create a new workplane at the origin, on the XY plane.
- Create a square profile centered on the origin
  - Use the **Rectangle** tool to sketch a 300 mm square
    - Enter -150, -150, then 150, 150 .



- RMB click on '/' in the tree and select **Create Empty Part**, name it 'pumpkin'.
- Pull the profile: **Create/Modify -> Pull**
  - Operation: Add Material
  - Mode: Linear
  - Direction: +W
  - Distance: 380
  - Click Done
- Pumpkin has taken the form of a large block and has been set **Active**.



- Hide (or delete) the workplane -- it's no longer needed, and its own lines could interfere with picking edges in the next step.
- Pumpkin should still be active.
- Fillet all of its edges: **Create/Modify -> Fillet**
  - Select all 12 edges
  - Enter radius value: 130



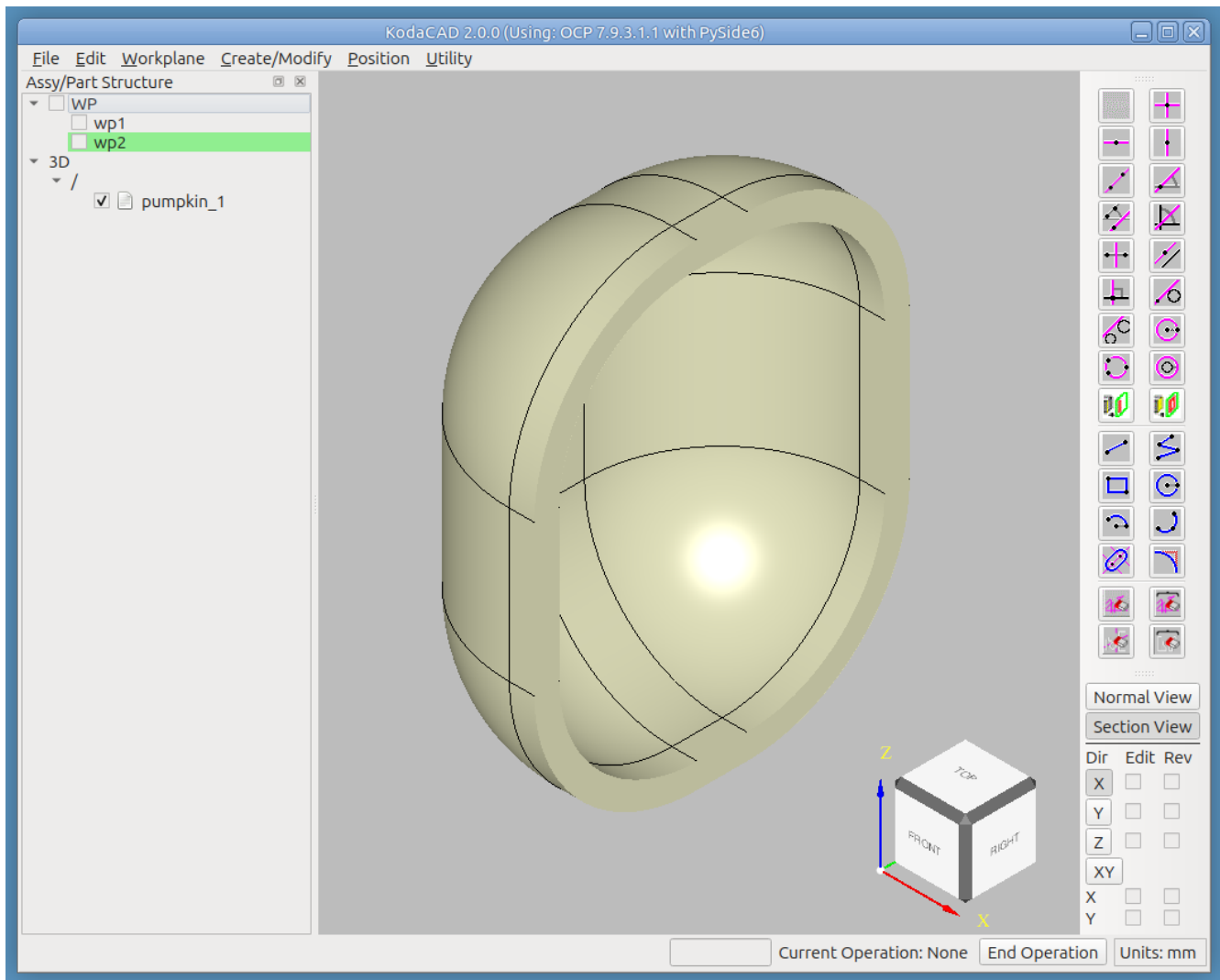
*Exercises: Rectangle-based pull into a simple box, then a large fillet operation turns that box into a rounded, pumpkin-like solid.*

## Step 5 -- Shell the pumpkin

- **Modify Active Part -> Shell:** click the small square top face, enter 20 for the shell thickness.

Worth knowing going in: that top face is entirely bounded by tangent fillets rather than genuine sharp edges shared with other faces, so Shell has no real opening to carve there -- the result comes out completely closed, not open at the top the way Shell normally produces. That's expected, not a bug: it's a known OCCT limitation around tangent (not sharp) face boundaries, confirmed independently in this project against both this pumpkin shape and a plain filleted bottle. It's also not a problem for this particular build, since the face will be cut through entirely in the next step regardless.

In order to verify that the pumpkin was properly shelled, click on the **Section View** button near the bottom of the toolbar.

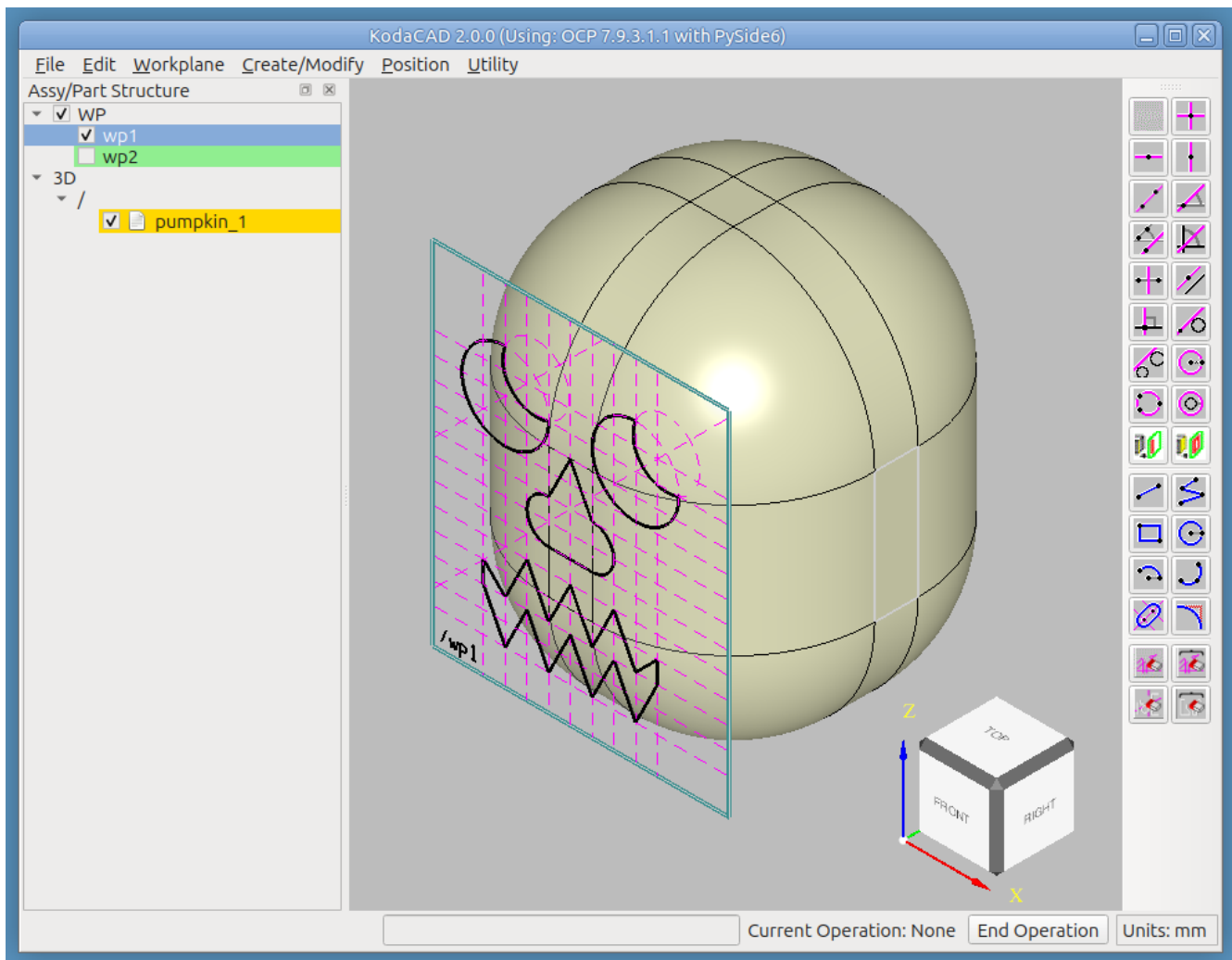


Initially, the clipped "face" of the section view will be in the X direction. Click on other directions to explore the way section views are displayed. Once in a section view, you can edit the location of the clipping plane. If you select **Normal View** again, then return to the section view just edited, the clipping plane you set will be remembered.

*Exercises: Shell on a form whose only candidate open face is entirely tangent-bounded, Section Views, and how to adjust the position of the clipping plane.*

## Step 6 -- Move workplane wp1 onto the front of the pumpkin

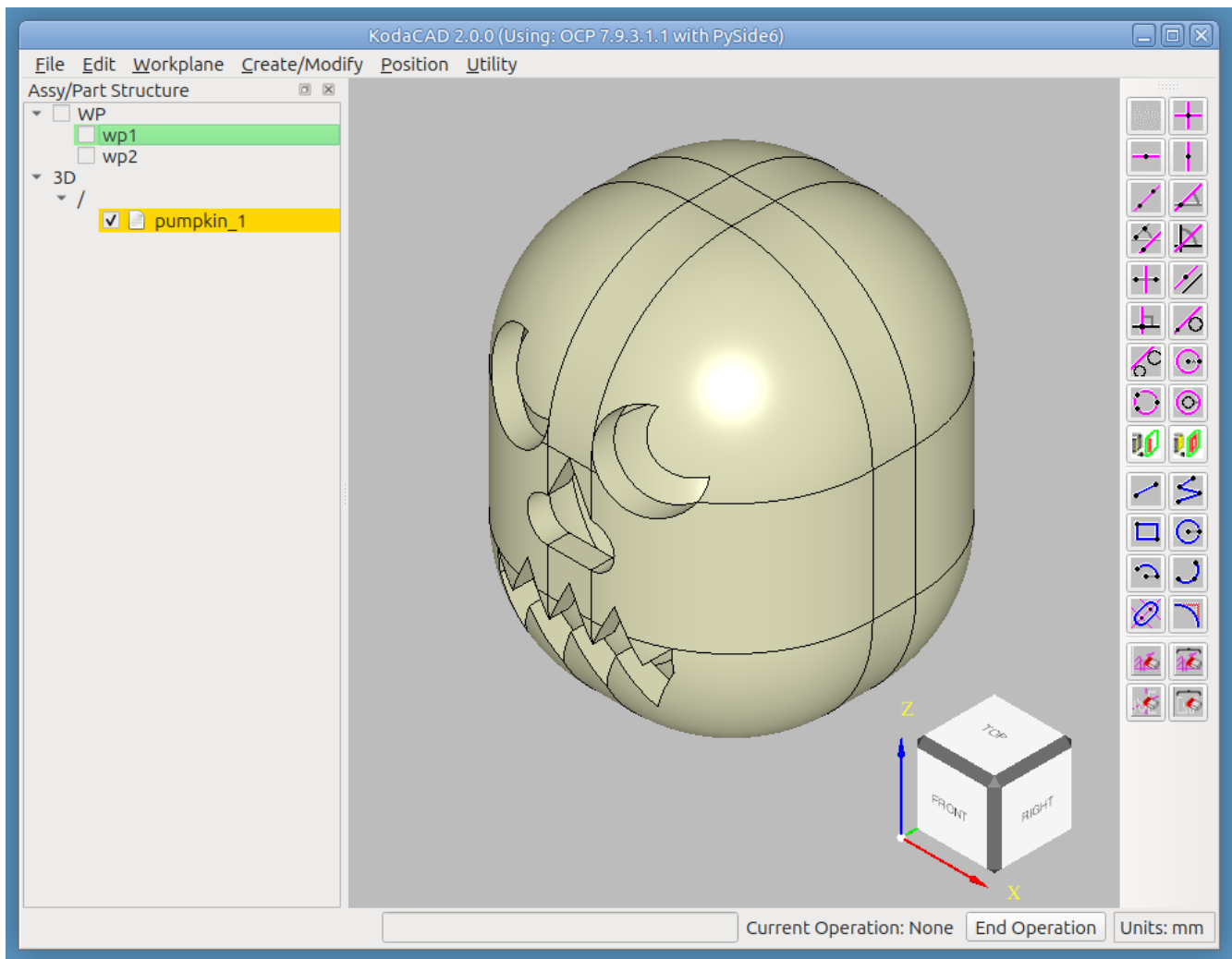
- Click on wp1 (with the face features) to select it, so that it is highlighted in blue.
  - In the menubar, click **Position -> Workplane**:
  - Click the front rectangular face (the workplane's origin will land at that face's geometric center)
  - Then click the right rectangular face for the +U direction.
- wp1 should now be positioned on the front face of the pumpkin.



*Exercises: Position -> Workplane -> On Face to move the workplane created in Steps 1-3, into position to carve the Jack-O-Lantern.*

## Step 7 -- Cut the face into the pumpkin

- Right click on wp1 in the tree and set it active. It will now be highlighted in green.
- Confirm the pumpkin part is set active (yellow) in the tree, and that the workplane with the face profiles is set active (green).
- **Create/Modify -> Pull**
  - Operation: **Remove Material**
  - Mode: Linear
  - Direction: **-W**
  - Distance: 150
  - Click ☒ **Done**
- You can now hide (or delete) the workplane.



*Exercises: Pull's multi-profile capability to remove material using four separate closed profiles.*

## Notes for whoever runs this next

- If any menu label, dialog control name, or button text has drifted from what's written here, that's worth fixing in this document directly -- catching that drift is exactly what this tutorial is for.
- See the OCC Bottle tutorial ([../occ-bottle/README.md](#)) for the pull-then-fillet technique this tutorial's own pumpkin body is built on, and for Pull's single-profile use as a point of comparison against this tutorial's multi-profile one.